Hierarchical Zooplankton Classification using CNN-RNN Sequence Modeling

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1. INTRODUCTION

Comprehensive monitoring of freshwater zooplankton requires automated classification of thousands of microscopy images for large-scale surveys. Standard flat classifiers treat all misclassifications equally, ignoring the fact that some errors are taxonomically minor (confusing two copepod subtypes) while others are major (confusing a copepod with a rotifer). In this work we design a classifier that respects biological hierarchy, providing partial credit for taxonomically consistent errors while maintaining high overall accuracy. We propose a CNN-RNN architecture that predicts complete taxonomy paths, encoding the organism category at multiple biological levels, rather than isolated leaf labels. This hierarchical framing improves interpretability, reduces misclassification penalties for within-group errors, and offers practical advantages for ecological applications that aggregate abundance by taxonomic group. This work draws inspiration from, and was motivatived by Raqui (2024) and Raqui and Haque (2024). Code is available at `https://github.com/ploewen/zooplankton`.

2. DATASET AND DATA PROCESSING

Dataset Description

Our dataset comprises 200,000+ preprocessed zooplankton particles extracted from Great Lakes microscopy mosaics, organized into 12 leaf classes across four major groups: Cladocera (Bosminidae, Daphnia), Copepoda (Calanoid, Cyclopoid, Harpacticoid, Nauplius_Copepod), Rotifera (Rotifer), and Non-biological (Bubbles, Exoskeleton, Fiber_Hairlike, Fiber_Squiggly, Plant_Matter). The raw dataset exhibits severe class imbalance, with some classes exceeding 180,000 images while others contain fewer than 1000 (see Figure 2). This motivates weighted sampling during training.

Exploratory data analysis reveals significant variability in image characteristics. Particle widths cluster at 0–200 pixels with outliers reaching 800+, heights range 0–400 pixels similarly, and aspect ratios exhibit right-skewed distributions peaked near 1.0. Image area is dominated by values below 20,000 px^2 with extreme outliers exceeding 100,000 px^2. This heterogeneity necessitates careful preprocessing to normalize inputs and reduce artifacts. The dataset follows a static 4-level taxonomy tree (see Figure 1) that provides natural partial-credit semantics where correct group classification counts as partial success even when leaf classification errs.

Preprocessing Pipeline

Particle Extraction. Raw data consists of large tiled microscopy mosaics paired with CSV files containing bounding box annotations. We extracted individual particles by reading bounding box coordinates (x, y, width, height) and cropping corresponding regions from each mosaic using bilinear interpolation.

Resizing and Normalization. Each extracted particle was resized to 64×64 pixels to balance computational efficiency against information preservation (most particles occupy $< 20,000$ px^2). We applied ImageNet normalization with mean $[0.485, 0.456, 0.406]$ and std $[0.229, 0.224, 0.225]$.

Dataset Organization. Images were stratified by class and randomly split into training (60%),

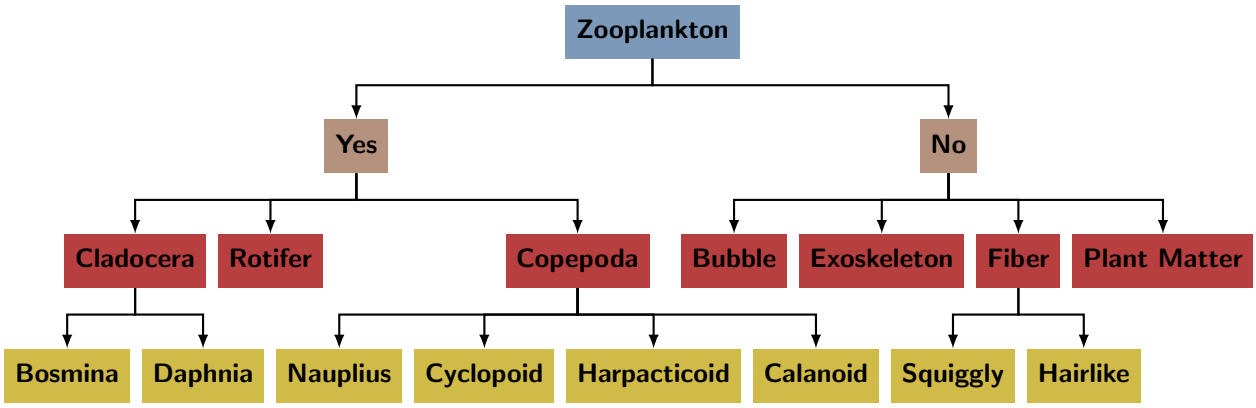

Figure 1: Hierarchical classification of biological and non-biological particles in zooplankton samples.

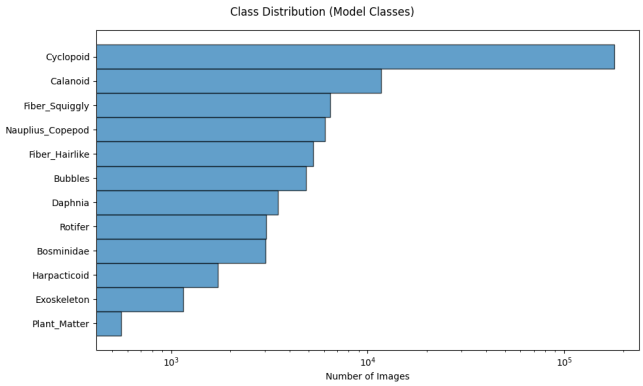

Figure 2: Class distribution (log scale) across the 12 model classes showing severe imbalance.

validation (20%), and test (20%) subsets, yielding approximately 230,000 preprocessed particles ready for model training.

Class Balancing for Training. To mitigate severe imbalance, we applied weighted class sampling during training. Each class received a weight inversely proportional to its frequency, and training batches were constructed by sampling with replacement according to these weights, following established practices (He and Garcia, 2009). Validation and test sets preserved the original class distribution for unbiased evaluation.

3. METHODOLOGY

Design Motivation: CNN-RNN for Hierarchical Classification Our choice of CNN-RNN architecture is naturally motivated by the structure of hierarchical classification. The problem can be viewed as a sequence of increasingly specific label predictions: first predicting whether the image is zooplankton or not, then refining to a specific family, and finally identifying the specific class. This sequential prediction naturally aligns with recurrent neural network architectures, which excel at modeling sequences and capturing dependencies across prediction steps. By formulating classification as sequence generation over a taxonomy tree rather than a flat multi-class problem, we leverage the LSTM's ability to maintain and update hidden state conditioned on prior predictions and visual features.

Simultaneously, convolutional neural networks are ideally suited to extract rich, spatially-aware visual representations from images. Rather than engineering domain-specific features, CNNs learn hierarchical feature maps that capture morphological patterns at multiple scales. ResNet18 (He et al., 2016) provides an efficient backbone for this task. Pretrained on ImageNet (Deng et al., 2009), ResNet18 offers strong representations while remaining computationally tractable for transfer learning on our moderately-sized dataset. Its skip connections enable stable training even with frozen weights, preserving the richness of pretrained features while avoiding catastrophic forgetting.

Hierarchical CNN-RNN Architecture Our model, denoted HierarchicalCNNRNN(θ), comprises three integrated components. The encoder uses a finetuned ResNet18 backbone, extracting a 512-dimensional feature vector $\mathbf{f} \in \mathbb{R}^{512}$ per image. A projection layer then maps encoder features to initialize LSTM hidden state via $\mathbf{h}_0, \mathbf{c}_0 = \text{Dense}_{512\to 2d}(\mathbf{f})$, where d is the LSTM hidden dimension, grounding sequence generation in visual

features.

As illustrated in Figure 3, the decoder consists of three modules: a token embedding layer converting vocabulary indices to dense embeddings $\mathbf{e}_t \in \mathbb{R}^{d_e}$ (where $d_e \in \{128, 256, 512\}$), an LSTM cell processing sequences autoregressively via $\mathbf{h}_t, \mathbf{c}_t = \text{LSTM}(\mathbf{e}_t, \mathbf{h}_{t-1}, \mathbf{c}_{t-1})$, and a linear output head mapping states to vocabulary logits $\mathbf{z}_t = \text{Linear}_{d \to V}(\mathbf{h}_t)$.

During training, we apply teacher forcing on image-sequence pairs. Teacher forcing is a training strategy where the model receives ground-truth tokens from the target sequence as input at each decoding step, rather than using its own previous predictions. Formally, given a target sequence $(y_1, y_2, \ldots, y_T)$, the decoder input at step t is the ground-truth token y_t, not the model's prediction $\hat{y}_{t-1}$. This approach provides two critical advantages for sequence-to-sequence learning. First, it stabilizes training by ensuring that early in optimization, when model predictions are poor, the decoder still receives meaningful context about the correct sequence structure. Without teacher forcing, errors early in the sequence would compound through subsequent timesteps (error propagation), causing gradient signal to vanish and training to diverge. Second, it provides strong token-level supervision at every decoding step, enabling the model to learn fine-grained hierarchical distinctions by observing correct parent predictions in sequence and learning to predict siblings conditioned on shared parental context.

At test time, we use autoregressive decoding without teacher forcing. We start with a START token and loop for up to 5 timesteps (maximum taxonomy depth), greedily selecting argmax predictions and feeding them back as input to the next step. This mimics deployment scenarios where ground-truth sequences are unavailable. We compute cross-entropy loss during training against all tokens except the first, masking padding tokens to prevent spurious gradients. The greedy decoding strategy is simple and interpretable, though beam search could recover higher-probability sequences at computational cost.

Training Configuration Our parameter optimization uses Adam with learning rate $\eta = 10^{-4}$, batch size 64, and gradient clipping at max norm 1.0. We employ weighted class sampling during training and early stopping on validation loss with patience 5 epochs and improvement threshold 0.003.

Loss Computation The loss is computed independently at each sequence position and flattened across the batch. For each sample and each timestep $t \in \{1, 2, \ldots, T\}$, we compute cross-entropy loss between the predicted logits $\mathbf{z}_t$ and the ground-truth token y_t

$$L_t = -\log p(y_t | \mathbf{z}_t)$$

We exclude the first token (START) from loss computation since it is given and deterministic. Padding tokens are masked to prevent spurious gradients. The final loss is the mean across all non-padding positions

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{B} \sum_{t=1}^{T_i} \mathbb{1}[\text{not padding at } (i,t)] \cdot L_t$$

where B is batch size, T_i is the sequence length for sample i, and N is the total number of non-padding positions.

Hyperparameter Tuning We tuned embedding size and hidden dimension over $\{128, 256, 512\}$. For each candidate configuration, we trained a fresh model on the training split and evaluated average token-level loss on the validation split. The configuration minimizing validation loss was selected as optimal and the best hyperparameters identified were embedding dimension $E = 512$, hidden dimension $H = 512$.

4. EXPERIMENTS AND RESULTS

Prediction Performance Figures 4–5 show that on the held-out test set, our model achieves 95% leaf-level accuracy (macro average precision 0.94, recall 0.95, F_1 0.95). Per-class analysis reveals strong performance on most categories. The best performing classes are Daphnia (Recall 0.99), Bubbles (0.99), Nauplius (0.98) while Harpacticoids show the lowest performance (Recall 0.88), likely due to morphological similarity to other small copepods and limited training samples in this rare class. Non-biological artifacts (Bubbles, Exoskeleton, Fibers) have very few errors, indicating clear visual separation from biological organisms.

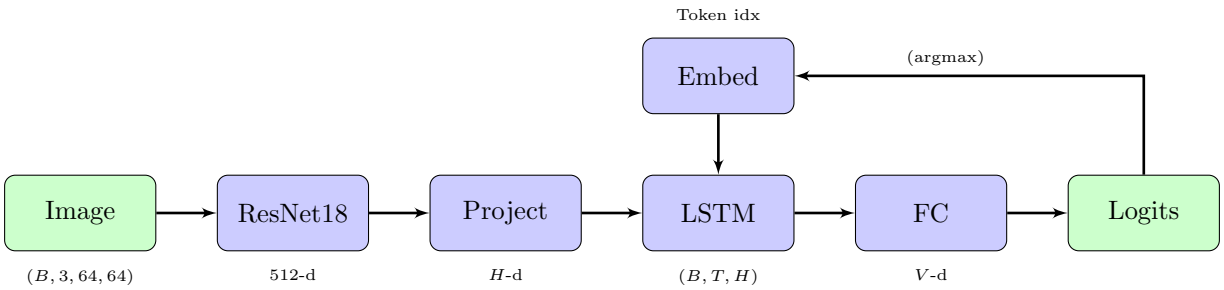

Figure 3: CNN-RNN Hierarchical Architecture. Image features from frozen ResNet18 are projected to initialize the LSTM decoder. Token embeddings feed the LSTM, which generates logits for next-token prediction.

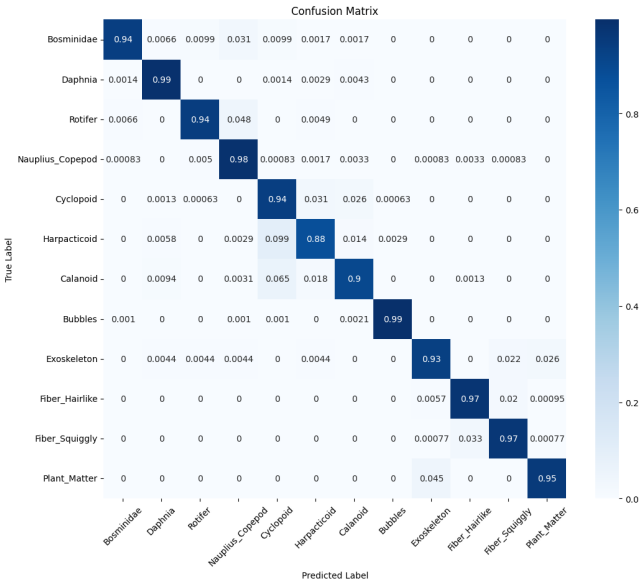

Figure 4: Leaf-level confusion matrix (95% accuracy). Darker blue indicates higher accuracy on diagonal. Most errors occur within biological groups (e.g., Harpacticoid vs Cyclopoid) rather than across major taxonomic boundaries.

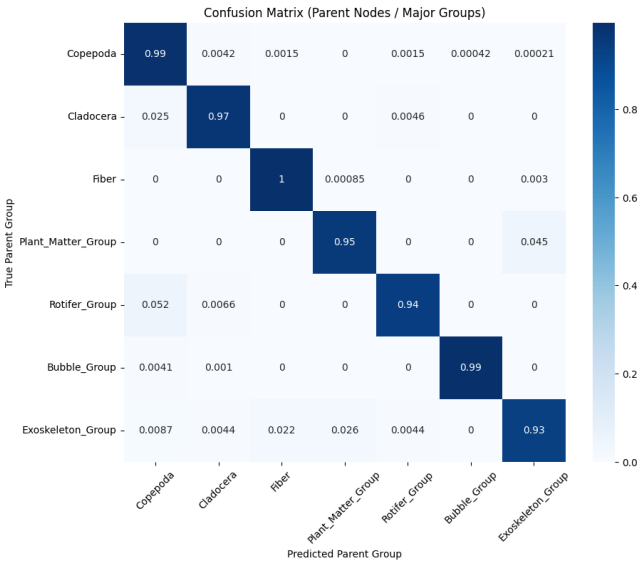

Figure 5: Parent-level confusion matrix (99% accuracy). Classification at the major group level demonstrates near-perfect taxonomic consistency.

Hierarchical Accuracy Advantage A key contribution is the ability to evaluate at coarser levels. Parent-level accuracy measures whether the predicted path correctly identifies the major group, regardless of leaf-class precision. Our model achieves 99.0% parent-level accuracy (7-class group classification) demonstrating that even when the model misclassifies a specific copepod subtype, it correctly identifies the Copepoda group 99.0% of the time.

This is a major advantage over flat classifiers where a Calanoid misclassified as Harpacticoid receives full error penalty, identical to misclassifying it as a Rotifer. In our hierarchical framework, the first error is penalized at the leaf level but correctly scored at the parent level, while the second error fails both levels. The confusion matrix reveals that most errors respect taxonomic boundaries. Although the model has relatively low performance when classifying Herpacticoids, all Copepod subtypes are rarely confused with non-biological categories or other major groups, validating the hierarchical approach's core premise that violations of hierarchy are rare and thus especially informative.

Error Analysis and Hierarchical Semantics Zooming into misclassification patterns, we observe that within-group confusions dominate. Harpacticoids are frequently misclassified as Cy-

clopoids or Calanoids but almost never as Rotifers. Cladocera (Bosminidae, Daphnia) are consistently preserved as a group. Non-biological particles have minimal error rates across all artifact categories. This pattern validates the hierarchical approach. Errors predominantly respect major taxonomic boundaries, meaning the hierarchy naturally encodes what the model reliably learns. For ecological applications aggregating by major group (e.g., total copepod abundance for a survey), the hierarchy reduces systematic bias that would arise from a flat classifier.

5. DISCUSSION

The hierarchical CNN-RNN architecture successfully integrates visual feature extraction with structured sequence prediction. Strong parent-level accuracy (99%) demonstrates that the model learns and respects biological taxonomy. The architecture naturally encodes the constraint that valid predictions follow the taxonomy tree, reducing pathological errors like predicting unrelated classes. Harpacticoid misclassifications persist despite hierarchical structure, suggesting that morphological similarity and limited training data are limiting factors independent of the classification framework. These rare classes benefit from shared representation through parent nodes but ultimately require additional visual discriminability.

The preprocessing pipeline directly influences model performance beyond logistical convenience. Standardizing to 64×64 balances computational efficiency against information preservation. Weighted class sampling effectively mitigated training instability due to severe imbalance. Without it, the model would overfit to frequent classes and underfit rare ones. The hierarchical structure provides additional regularization by requiring rare classes to be predicted through shared parent tokens, reducing per-class variance.

6. LIMITATIONS AND FUTURE DIRECTIONS

The taxonomy is predefined and biologically motivated rather than learned from data. Data-driven hierarchy discovery could identify informative groupings, though at increased complexity. Evaluation on 12 classes from a single imaging setup means generalization to other plankton taxa or imaging modalities remains unexplored. Our greedy decoding, while interpretable, may miss higher-probability sequences that beam search would recover. Detailed analysis of inference latency and deployment constraints was not performed. Future work should explore learned hierarchies, ensemble and multi-modal fusion, generalization to additional taxa and imaging systems, uncertainty quantification via Bayesian variants, direct comparison with equivalent flat classifiers, and ecological validation against expert-classified field surveys.

7. CONCLUSION

We present a hierarchical CNN-RNN architecture for automated zooplankton classification that predicts taxonomy sequences instead of flat labels. By respecting biological taxonomy, the model achieves 95% leaf-level accuracy and 99% parent-group accuracy, demonstrating that hierarchical structure naturally constrains predictions to be taxonomically consistent. The approach improves interpretability, provides partial correctness semantics valuable for ecological aggregation, and offers practical advantages for Great Lakes monitoring. Our work bridges deep learning and ecological science, leveraging domain knowledge to improve both performance and real-world utility.

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APPENDIX

A. HYPERPARAMETER TUNING METHODOLOGY

Grid search explored embedding size and hidden dimension over $\{128, 256, 512\}$. For each configuration, we trained a fresh model on the training split and evaluated token-level loss on the validation split. The configuration minimizing validation loss was selected. This full reinitialization prevented information leakage across trials.

B. MODEL AND DATA ASSUMPTIONS

The architecture assumes:

- Fixed-depth taxonomy (sequences pad to maximum length ≤ 5)
- Pixels contain sufficient morphological information (64×64 resolution adequate)
- Greedy decoding suffices (beam search not explored)

The dataset assumes:

- Stratified splits preserve class distributions in validation and test
- Weighted sampling during training adequately addresses class imbalance
- Bounding box annotations are accurate
- The 3-level biological taxonomy correctly structures the domain

C. MODEL LIMITATIONS

- **Fixed hierarchy:** The taxonomy is predefined; learned hierarchies could be more data-adaptive
- **Single imaging setup:** Evaluation on one imaging apparatus limits generalization claims
- **Class imbalance impact:** Despite weighting, rare classes (e.g., Plant_Matter with 554 samples) provide weak supervision
- **Greedy decoding:** Autoregressive sampling is simple but may miss higher-probability paths
- **Frozen vs. fine-tuned trade-off:** Fine-tuning improves accuracy but risks overfitting; ablation shows frozen baseline is substantially worse

D. ABLATION STUDY: FROZEN VS. FINE-TUNED RESNET18

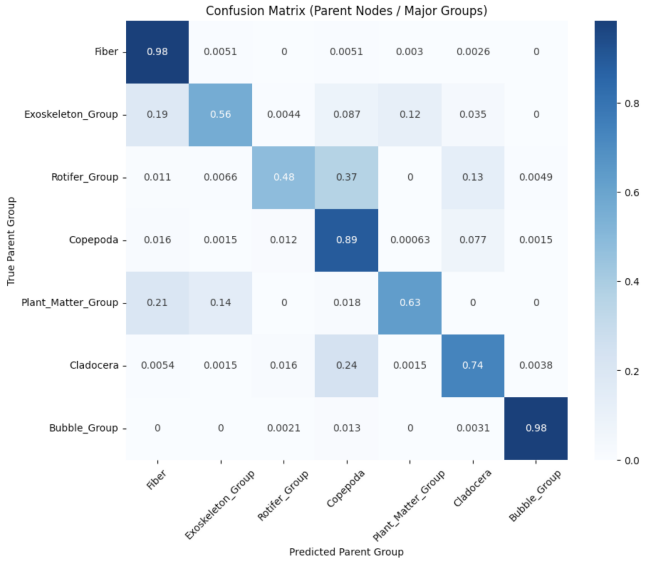
To evaluate the necessity of fine-tuning the ResNet18 encoder, we compared our proposed fine-tuned model against a variant with frozen ResNet18 weights. The frozen baseline maintained all ImageNet-pretrained weights fixed throughout training, with only the projection layer, LSTM, and output head being trainable.

Results demonstrate that fine-tuning is critical for performance. The frozen ResNet18 variant achieved only 75% leaf-level accuracy and 87% parent-level accuracy, compared to our full model's 95% leaf-level and 99% parent-level accuracy. This represents a 26.7% relative improvement at the leaf level and 13.8% at the parent level. The frozen model's degraded performance suggests that

while ImageNet features provide a useful initialization, the ImageNet distribution differs sufficiently from zooplankton microscopy that adaptation of the visual encoder is necessary to fully exploit the zooplankton-specific morphological patterns.

Figures 6a and 6b show the confusion matrices for the frozen model, revealing reduced per-class discrimination and higher off-diagonal error rates compared to the fine-tuned model. This ablation validates the design choice to fine-tune the ResNet18 encoder and demonstrates that hierarchical structure alone cannot compensate for domain-mismatched visual representations.

(a) Leaf-level confusion matrix for the frozen ResNet18 variant (75% accuracy). Compared to the fine-tuned model (95% accuracy shown in main paper), the frozen variant exhibits increased confusion across classes, particularly in rare copepod subtypes and similar-looking taxa.



(b) Parent-level confusion matrix for the frozen ResNet18 variant (87% accuracy). While parent-level accuracy remains high, the frozen model shows degraded performance compared to the fine-tuned variant (99% accuracy).

Figure 6: Frozen ResNet18 ablation study: confusion matrices for the variant with frozen encoder weights. Left: leaf-level performance (75.0%) compared to fine-tuned (90.0%). Right: parent-level performance (87.0%) compared to fine-tuned (98.1%). Visual fine-tuning proves critical for domain-specific performance.